

Problem Statement No. 1: Nutrition Agent

The Challenge

Good nutrition is the foundation of health, but individuals often struggle with diet planning, portion control, and tracking nutritional intake. Information about healthy diets, disease-specific nutrition, and food composition is scattered across unreliable sources. Without personalized, real-time guidance, many fail to maintain balanced diets or make informed food choices.

The Objective

Build an AI-powered Nutrition Agent that delivers personalized dietary insights and guidance for individuals, families, and health professionals. The solution should:

- 1 **Personalized Diet Plans:** Recommend meals based on age, health conditions, allergies, cultural preferences, and fitness goals.
- 2 **Food Data RAG Retrieval:** Fetch and summarize nutritional values (calories, macros, vitamins) from reliable sources.
- 3 **Diet Tracking & Feedback:** Allow users to log meals via text, images, or voice, and provide instant nutritional analysis.
- 4 **Preventive Health Focus:** Suggest diets for chronic disease management (e.g., diabetes-friendly or heart-healthy plans).
- 5 **Multi-Agent System:** Nutrition Knowledge Agent (fetches nutritional data); Diet Recommendation Agent (personalized meal plans); Health Advisory Agent (preventive health suggestions); Food Log & Feedback Agent (meal tracking + analysis).
- 6 **Visualization Dashboard:** Show daily/weekly nutrient intake, deficiencies, and progress toward health goals.

Tech Stack (Mandatory)

IBM Langflow / IBM Orchestrate + IBM Models